CANDIDATE ELIMINATION

Terms used :-

→ **Concept Learning** — concept learning is basically learning task of the machine (learn by Train Data).

→ **General Hypothesis** :- Not specifying features to learn the machine

→ G = { '?', '?', '?', '?' ... } No. of attributes.

→ **Specific Hypothesis** — Specifying features to learn machine (specific feature).

S = { 'pi', 'pi', 'pi', ... } : No. of pi depends on no. of attributes.

Version Space :- It is intermediate of General Hypothesis and Specific Hypothesis. It not ~~no~~ only just written one hypothesis but a set of all possible hypothesis based on training data set.

Concept Learning
- ↳ General Hypothesis
- ↳ Specific Hypothesis
- ↳ Version Space

Learned Version Space by Candidate Elimination Algo. for Given Data Set is —

Eg

Example	Sky	Air Temp	Humidity	Wind	Water	For Caste	Enjoy Sport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

The Algorithm incrementally build the version space given a hypothesis space H and a set E of examples.

It is used to find the set of consistent hypothesis that is Version Space.

→ Uses Version Space

→ Consider both +ve and −ve results.

→ We have both specific and general hypothesis.

→ For a +ve example :-
We tend to generalize Specific hypothesis [FIND-S].

→ For a −ve example :-
We tend to make general hypothesis to more specific

$S = \{\phi, \phi, \phi, \ldots \phi\}$

$G = \{?, ?, ? \ldots ?\}$

$S = \{\phi, \phi, \phi, \ldots \phi\}$

↓

↑

$G = \{?, ?, ?, \ldots ?\}$

(Bottom to top)

We achieve more specificity we encounter a −ve example

ALGORITHM (a) Load Data

1. Intialize G and S as most general and Specific hypothesis.
   (b) for each training example.
2. For each example e :
   if e is +ve :
   → Make specific hypothesis more general [FIND-S]
   else : → Make generalize hypothesis more specific

Example ①

Concept : Days on which person enjoys Sport.

yes → +ve
NO → −ve

S.1 : Load Data Set

S2 : Intialize General Hypothesis and Specific Hypothesis.

S3 : ~~If Example~~ for each training example

S4 : If example is +ve example :
if attribute-value == Hypothesis-value :
Do nothing

else :
replace attribute value with '?' ?
{ basically ~~both~~ Generalizing it }.

S5 : If example is -ve example :
Make Generalize hypothesis more specific

}

}

example
Algorithmic Steps :-

Initially : $G_0 = [[?,?,?,?,?,?],[?,?,?,?,?,?],[?,?,?,?,?,?],[?,?,?,?,?,?],[?,?,?,?,?,?],[?,?,?,?,?,?]]$
S.1 Most Generic Boundary

$S_0 = [\phi, \phi, \phi, \phi, \phi, \phi]$ most specific Boundary

S.2 $S_1 = \{$'Sunny', 'Warm', 'Normal', 'Warm', 'Same'$\}$

$G_1 = \{?,?,?,?,?,?]$

first example is +ve, hypothesis at specific boundary is inconsistent, hence we extend specific boundary & the hypothesis at the generic boundary is Consistant so we retain it.

S.3 $S_2 = \{$Sunny, 'Warm', ?, 'Warm', 'Same'$\}$

$G_2 = (?,?,?,?,?)$

2nd is +ve example, again hypothesis at specific boundary is inconsistent so we extend specific boundary & hypothesis at generic boundary is consistent so we retain it.

S.4 S_3: (Sunny, Warm, ?, Strong, Warm, Same)

G_3: (Sunny, ?, ?, ?, ?, ?) (?, Warm, ?, ?, ?, ?).

(?, ?, ?, ?, ?, Same)

3. Third example is -ve, the hypothesis at the specific boundary is consistent, hence we retain it and hypothesis at generic boundary is inconsistent hence we write all consistent hypotheses by removing one '?' at a time.

S.5 fourth is +ve, hypothesis at the specific boundary is inconsistent hence we extend specific boundary. & consistent hypothesis at generic boundary are retained

S_4: (Sunny, Warm, ?, Strong, ?, ?)

G_4: (Sunny, ?, ?, ?, ?, ?) (?, Warm, ?, ?, ?, ?)

Version Space by Candidate Elimination Algo for Given Data set is:-

S: {<Sunny, Warm, ?, Strong, ?, ?>}

<Sunny, ?, ?, Strong, ?, ?> <Sunny, Warm, ?, ?, ?, ?> <?, Warm, ?, Strong, ?, ?>

G: {<Sunny, ?, ?, ?, ?, ?>, <?, Warm, ?, ?, ?, ?>}